Fairness-Aware Two-Stage Hybrid Sensing Method in Vehicular Crowdsensing

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***Abstract*—By utilizing on-board sensors and computing resources in intelligent vehicles, vehicular crowdsensing can collect a series of sensing data. Typically, sensing vehicles can be divided into opportunistic vehicles with fixed trajectories and participatory vehicles with changeable trajectories. Therefore, to complete sensing tasks more effectively, how to combine the advantages of the mobility characteristics of the two vehicles is a challenging problem. To solve this problem, this paper innovatively proposes a joint scheduling and incentive-driven two-stage hybrid sensing method. Specifically, the method is divided into two stages: opportunistic vehicle selection and participatory vehicle scheduling. In particular, both types of vehicles are managed through the Crowd Sensing Platform (CSP). For the first stage, this paper proposes a reverse auction-based incentive mechanism to select the lowest-cost set of vehicles to complete sensing tasks. This mechanism mainly consists of two steps: winning vehicle selection and reward payment. It is also verified that the proposed mechanism can ensure the individual rationality and truthfulness of opportunistic vehicles. For the second stage, based on the first-stage sensing results, this paper proposes a Soft Actor-Critic (SAC) based approach to scheduling participatory vehicle trajectories to complete sensing tasks. In addition, this paper also considers sensing fairness to ensure the balance of sensing task completion in different sub-regions. Through the two-stage hybrid sensing method, this paper aims to minimize the CSP overhead while ensuring sensing fairness. Finally, extensive evaluation results based on Roma taxi data sets demonstrate that the proposed method works effectively and outperforms other benchmark schemes in different working scenarios.**

***Index Terms*—Fairness, hybrid vehicle sensing, reverse auction, soft actor - critic (SAC), vehicular crowdsensing.**

I. Introduction

Recently, with the rapid development communication technology and sensor technology, smart cities have entered the era of the intelligent Internet-of-Everything (IoE) [1], [2]. Fortunately, Mobile Crowd Sensing (MCS) has been applied to constructing smart cities [3]. Compared to traditional sensor networks, MCS boasts several advantages, such as low data collection costs, ease of device maintenance, and scalable system architecture [4]. In addition, ubiquitous vehicles have played an important role in smart cities. As vehicles have become networked and intelligent, they have transformed from a convenient means of transportation into a powerful platform for mobile sensing, computing and storage. This has led to the emergence of Vehicular Crowd Sensing (VCS), in which vehicles serve as the basic sensing units [5]. It supports Internet-of-Vehicles (IoV) services, enabling coordination between humans, vehicles, and the environment in urban road traffic scenarios [6]. In particular, Vehicular Ad Hoc Networks (VANETs) realizes a network structure without central scheduling through self-organized communication between vehicles. Along with 5G/6G in supporting high speed interconnection with infrastructure and core network, VANETs apply short-range communication in a flexible and distributed interconnection manner. Meanwhile, vehicles are mobile and flexibly operated, and using VANETs can support large-scale natural and social sensing needs in smart cities.

In VCS, according to the way vehicles participate in sensing tasks, they can be divided into opportunistic vehicles and participatory vehicles [7], [8]. Meanwhile, the sensing involving opportunistic vehicles is called opportunistic sensing, and the sensing involving participatory vehicles is called participatory sensing [9], [10]. Fig. 1 shows a schematic diagram of the two types of vehicle sensing. Specifically, opportunistic vehicles do not change their original trajectories and rely on their daily driving paths to perform sensing tasks [11]. In contrast, the participatory vehicle can complete the task by changing its initial trajectory based on the task location.

A specific example is given below to illustrate how opportunistic vehicles and participatory vehicles participate in sensing tasks, and then introduces the method proposed in this paper. For example, For-Hire Vehicles (FHVs) in the city, including taxis and online ride-hailing vehicles, can serve as sensing entities to complete sensing tasks. For FHV, according to whether there are order tasks, it can be divided into two states: idle and passenger-carrying, which can also correspond to participatory vehicles and

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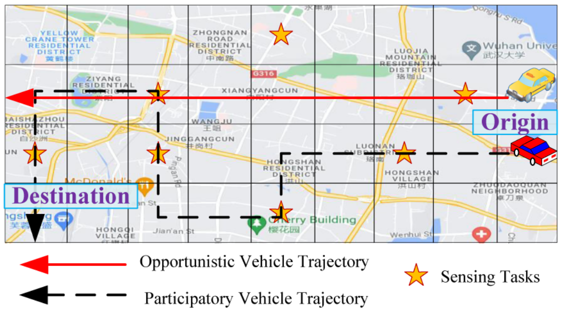


Fig. 1. The opportunistic sensing and participatory sensing.

opportunistic vehicles. Here, the passenger-carrying FHV needs to take passengers to their destinations according to a prescribed route, and its route cannot be changed, so it is regarded as an opportunistic vehicle. It only completes the sensing tasks assigned on the existing driving trajectory. However, due to the limited coverage of fixed routes of passenger-carrying FHVs, there will be situations where some regions complete multiple sensing tasks while some regions have not yet been sensed. On the contrary, idle FHVs can overcome this disadvantage. Here, the idle FHV does not have a fixed driving trajectory, so it can actively change its trajectory according to the location of the sensing task and is regarded as a participatory vehicle. In particular, it can travel to regions where sensing tasks are less accomplished to compensate for the lack of opportunistic vehicles. Compared with opportunistic vehicles, in addition to receiving rewards for completing tasks, corresponding detour rewards will also be generated. Here, the detour reward is a floating reward related to the detour distance. However, participatory vehicles may travel long distances to complete tasks for higher rewards, and this inevitably increases the overtravel of the platform. Therefore, how to reasonably utilize the advantages of the two types of vehicles to better complete the sensing task is a question we need to consider.

Typically, performing sensing tasks requires additional time or economic costs for vehicles. Since most vehicles are selfish and rational, it is often unfeasible to drive them to perform sensing tasks without rewards. Therefore, an appropriate reward mechanism is needed to motivate vehicles to participate in sensing tasks [12], [13]. Here, the incentive mechanism is mainly introduced from the economics perspective and gradually applied to crowdsensing. Currently, commonly used methods include auction theory, game theory, contract theory, etc. Then, since participatory vehicles do not have predetermined driving trajectories, the platform is required to serve as a decision-maker to schedule their trajectories. Fortunately, Deep Reinforcement Learning (DRL) methods can determine real-time actions through real-time interaction with the environment [14], [15]. Therefore, this method can provide real-time optimal decisions for participatory vehicles and can adapt to large-scale complex scenarios. In addition, considering the differences in sensing tasks completed by these two types of vehicles, it is necessary to address the issue of sensing fairness in different sub-regions. Here, the fairness index can be introduced into the two types of vehicle sensing processes to solve this problem.

In summary, combining the advantages of both types of vehicles becomes the potential for our attention in VCS. Previous works inadequately considered combining the mobility characteristics of the two types of vehicles to exploit their advantages to ensure sensing fairness [13], [16]. To fill this gap, taking the different mobility characteristics of the two types of sensing vehicles as the starting point, this paper studies an innovative two-stage hybrid sensing method in vehicular crowdsensing. Based on the different ways in which vehicles complete sensing tasks, we combine the incentive characteristics of reverse auctions and the dynamic scheduling of DRL. In this way, the advantages of both types of vehicles can be effectively combined to better complete sensing tasks. In addition, the completion of sensing tasks in different sub-regions also needs attention. Thus, this paper also takes sensing fairness into account to ensure the balance in the number of sensing task completions in different sub-regions.

The main contributions of this paper are summarized as follows:

1) Our paper takes the mobility of two types of vehicles as the starting point, makes full use of the characteristics of vehicles, and pioneeringly proposes a two-stage hybrid sensing method in vehicular crowdsensing scenarios. The method is accomplished by selecting an optimal set of opportunistic vehicles and scheduling optimal driving trajectories of participatory vehicles. Specifically, in the first stage, we propose a reverse auction-based opportunistic vehicle selection method to select the optimal set of opportunistic vehicles to complete the sensing task. Based on the results of the first stage, we propose a Soft Actor-Critic (SAC)-based participatory vehicle scheduling method in the second stage to schedule the participatory vehicle trajectory to further complete the sensing task.
2) Along with the effort in integrating two types of vehicles for task sensing, we further consider the sensing fairness index, which will avoid the problem of sensing imbalance among different sub-regions. Then, we introduce the sensing fairness index into the optimization objective in order to minimize the overhead of Crowd Sensing Platform (CSP) through the two-stage hybrid sensing method. Specifically, in the first stage, the optimization objective is mainly achieved by selecting the lowest-cost opportunistic vehicle set. In the second stage, the optimization objective is further achieved by scheduling opportunistic vehicles to improve sensing fairness. In particular, we set the fairness index as the reward function of the DRL method in the second stage. Through the two-stage hybrid sensing method, the mobility advantages of both types of vehicles can be effectively combined.

The rest of this paper is organized as follows. Section II introduces related works. Section III outlines the system model related to this paper and proposes optimization problems. Section IV presents the method for solving the opportunistic vehicle selection problem in the first stage of the sensing process. Section V presents the solution method for the participatory vehicle scheduling problem in the second stage of the sensing process. Section VI describes the simulation experiments and analyses the performance results. Finally, Section VII concludes this paper.

II. Related Works

A. Vehicular Crowdsensing

In recent years, many studies have considered the vehicle recruitment and selection issues in vehicular crowdsensing. For example, Hu et al. [17] proposed a vehicle recruitment method based on variable duration. The selected vehicle and service duration are determined by considering vehicle uncertainty trajectories under limited budget conditions. Then, the authors designed a two-stage recruitment algorithm to maximize the use of sensing resources. In addition, some other studies focus on vehicle path planning and selection issues. For example, Lai et al. [18] studied the problems including vehicle recruitment, candidate route analysis, and route selection. First, they analyzed vehicle preferences and interests based on the historical vehicle trajectory information. Then, they addressed the maximum weighted path detection problem based on the minimum interference-aware strategy. Finally, the path with the least disruption to vehicle travel and the greatest benefit was selected.

Recently, the latest research has begun to focus on opportunistic vehicles and participatory vehicles. Authors in [10] divided taxis into multiple types according to their different states, which can correspond to participatory vehicles and opportunistic vehicles. Then a corresponding incentive mechanism is proposed to encourage taxis in different states to participate in sensing tasks. The goal is to increase system utility and achieve higher revenue for taxis. Fu et al. [7] proposed a hybrid vehicle recruitment scheme based on deep learning Here, suitable vehicles are recruited by predicting vehicle trajectories and task density to maximize the task completion rate. First, an opportunistic vehicle recruitment method based on deep learning is proposed, and then a participatory vehicle recruitment method based on sensing density is proposed. Our paper is significantly different from and goes beyond the above-mentioned studies, as detailed in Section II-D.

B. Reverse Auction-Based Incentive Mechanisms

In view of the individual rationality and selfishness of the vehicle, it is difficult to actively complete the sensing task without rewards. Therefore, an incentive mechanism needs to be introduced to prompt the vehicle to complete the sensing tasks, such as game theory, auction theory, contract theory, etc. Here, the reverse auction is one of the feasible methods [19].

In recent years, the reverse auction method has been applied in vehicle crowdsensing. For instance, Fan et al. [16] proposed a method called Hector for joint trajectory scheduling and incentive mechanism of vehicle sensing network. Here, vehicle trajectory is mapped to a set of spatiotemporal constraints to reduce the complexity of problem-solving. Meanwhile, in order to achieve wider task coverage, authors designed a reverse auction-based incentive mechanism to encourage vehicles to detour. Gao et al. [20] studied an incentive mechanism based on vehicle non-determinism. To encourage vehicles to participate in sensing tasks, they established a reverse auction model between the platform and vehicles. In addition, by considering the sensing data quality of vehicles, the authors proposed corresponding winner selection methods and payment determination methods. Chen et al. [21] proposed an incentive mechanism based on timeliness. Then, a reverse auction model between service requesters and vehicles was established. By transforming the auction problem into a non-monotone sub-module maximization problem, the utility of the requester under budget constraints was improved.

C. DRL for Crowdsensing

With the continuous development of vehicular crowdsensing systems, new technologies are needed to solve the challenges brought by the increasing system complexity. The DRL method shows its unique advantages in solving complex dynamic problems. Zhao et al. [22] studied non-cooperative vehicle sensing methods. Here, the unit completion cost of sensing tasks is dynamically determined, and relevant vehicles were motivated to participate in sensing tasks through the social attributes of vehicles. To ensure the long-term optimality of decision-making, Xu et al. [23] combined generative adversarial networks and DRL to solve the sensing task allocation problem. Here, actions can be flexibly and adaptively adjusted according to the environment. Furthermore, the authors integrated the generative adversarial network into the DRL training process. Then, the dual deep Q-learning method was applied in the training stage, thus ensuring the optimal strategy in the sensing decision-making stage.

With the development of intelligent vehicles, recent research also considers distributed sensing scenarios, where vehicles can also serve as decision-making agents [24]. Ding et al. [25] proposed a graph convolution collaboration-based multi-agent RL method, which can assist taxis in distributed decision-making and collaborative optimization. Here, vehicle cooperation was promoted by introducing a trust distribution method. Meanwhile, by combining with the graph convolution network, the spatial features of the complex road network were collected. In this way, the vehicle's decision-making actions can be further guided. Liu et al. [26] proposed a distributed multi-agent DRL method for energy saving and distributed unmanned vehicle scheduling. This method extracts features through convolutional neural networks to generate real-time actions. This can better guide the trajectory of autonomous vehicles. In addition, better exploration strategies can be achieved by using long short-term memory networks and distributed priority experience replay. This will reduce the energy consumption of autonomous vehicles.

D. Motivation

Our paper goes beyond the state-of-the-art in terms of adaptability and flexibility. Most existing work consider completing sensing tasks from the perspective of a single type of vehicle. This does not fully exploit the advantages of the vehicle's mobility characteristics, nor does it avoid its shortcomings. Therefore,

TABLE I
SYMBOLS AND DEFINITIONS

Symbol	Definition
$u_m \in \mathbf{U}, v_n \in \mathbf{V}, l \in L$	opportunistic vehicles set, participatory vehicles set, and sub-region grids set
s_m^h	The number of grids that opportunistic vehicle u_m passed
c_m^l, c_n^l	The unit cost of vehicle u_m or v_n performing sensing task
$x_m = \{0, 1\}$	Whether the opportunistic vehicle u_m is selected to perform sensing task
b_m	The unit bid price of opportunistic vehicle u_m
Dis_n	The actual driving distance of participatory vehicle v_n
D_n^{max}	The planned driving distance of participatory vehicle v_n
c_d	The unit detour cost of participatory vehicle v_n
h_n	The current number of sensing tasks completed by v_n
$count_l^u, count_l^v$	The total number of sensing counts of u_m or v_n in sub-region grid l
η	The number of sensing counts in each sub-region grids should not exceed η
g_n	The max number of sensing tasks completed by v_n

based on the mobility characteristics of opportunistic vehicles and participatory vehicles, we combine the advantages of both to complete sensing tasks more flexibly. Then, based on the different ways in which the two types of vehicles complete sensing tasks, we combine the incentive properties of reverse auctions and the dynamic adaptability of DRL. This combination can be matched with our two-stage hybrid sensing method, showing unique adaptability. Furthermore, we introduce sensing fairness in vehicular crowdsensing scenarios, which is not covered by the existing studies.

III. SYSTEM MODEL

This section will introduce the basic definitions of the two types of vehicles, and give the corresponding problem formulation, then propose the optimization objectives of this paper. Table I summarizes the key notation used in this paper.

A. Model Definition

1) System Model Overview: This paper proposes a two-stage hybrid sensing method applied to vehicular crowdsensing scenarios. We consider a scenario consisting of a set of opportunistic vehicles, a set of participatory vehicles, and the Crowd Sensing Platform (CSP). CSP is responsible for issuing sensing tasks, while managing and scheduling the above two types of vehicles to complete sensing tasks. In this paper, the sensing task is defined as collecting specific data (e.g., images or videos) in a target region within a specific time period. This paper selects a rectangular target region and divides it into sub-region grids of the same size, represented by $\mathbf{L} = \{1, \ldots, L\}$. Each sub-region grid has a sensing task that needs to be completed. We assume that there are M opportunistic vehicles in the rectangular target region, expressed as $\mathbf{U} = \{u_1, \ldots, u_m, \ldots, u_M\}$. There are N participatory vehicles, represented by $\mathbf{V} = \{v_1, \ldots, v_n, \ldots, v_N\}$. When a vehicle passes through the current sub-region grid, it represents the completion of a sensing task. For the sensing task in each sub-region, it is stipulated that the number of data collected by vehicles should not exceed η counts. That is, the completion of sensing tasks in each sub-region cannot exceed η counts.

2) Hybrid Sensing Overview: Fig. 2 shows our proposed two-stage hybrid sensing method. Specifically, Fig. 2(a) shows the sensing process of opportunistic vehicles in the first stage. In this process, opportunistic vehicles have fixed trajectories, and the CSP selects the optimal set of opportunistic vehicles to complete the sensing task through the reverse auction method. After the first stage of sensing process, different sub-region grids completed different counts of sensing tasks. Fig. 2(b) shows the sensing process of the participatory vehicle in the second stage. Here, the blue color of the sub-region grid represents the completion counts of sensing tasks in this sub-region in the first stage. The darker the blue, the more counts of sensing tasks have been completed in this sub-region. During the second stage, the trajectory of the participatory vehicles can be changed based on demand, especially the lighter blue sub-regions. Therefore, based on the sensing results of the first stage, the CSP uses the DRL method to schedule participatory vehicle trajectories to further complete the sensing tasks.

3) Sensing Fairness Overview: Here, we also introduce the definition of sensing fairness to ensure the sensing fairness of different sub-regions. Here, the target region is divided into multiple sub-region grids, and multiple sensing tasks can be performed within each sub-region grid. Therefore, CSP needs to ensure the balance of the number of sensing tasks completed among different sub-region grids. That is, in Fig. 2, the shade of blue in each sub-region grid needs to be as consistent as possible. This will avoid the situation where some sub-regions have multiple sensing tasks completed while some sub-regions have not yet been sensed. In particular, we consider sensing fairness during two stages of the hybrid sensing processes. Our goal is to improve the sensing coverage, maintain the sensing fairness of different sub-region grids, and minimize the overhead of CSP through the two-stage hybrid sensing process.

4) Other Basic Definitions: In addition, the entire sensing window is defined as time D, where each sensing task is valid within time D. Moreover, the vehicle can complete the sensing task only if it arrives within the time range. This paper assumes that the time window is $[0, D]$ and is divided into two equal parts to match the two-stage sensing process. Specifically, the sensing time of opportunistic vehicles in the first stage is $[0, \frac{1}{2}D]$, and the sensing time of participatory vehicles in the second stage is represented as $[\frac{1}{2}D, D]$. In the second stage of the participatory vehicle sensing process, time can also be discretized into T time slots, expressed as $\{1, \ldots, t, \ldots, T\}$. In the opportunistic sensing scenario, the platform makes decisions offline, and the opportunistic vehicles perform tasks, so time slot division is not considered. In the participatory sensing scenario, time slots need to be divided to make real-time decisions for each participatory vehicle. In addition, the total budget for CSP to perform sensing tasks is set to B. That is, it is necessary to ensure that the vehicle's cost of performing the sensing task is less than B.

5) The Impact of Communication Network: In our scenario, the two types of vehicles mainly establish connections with the CSP through the cellular network, accept the control of the CSP and transmit sensing data. There are similarities and differences

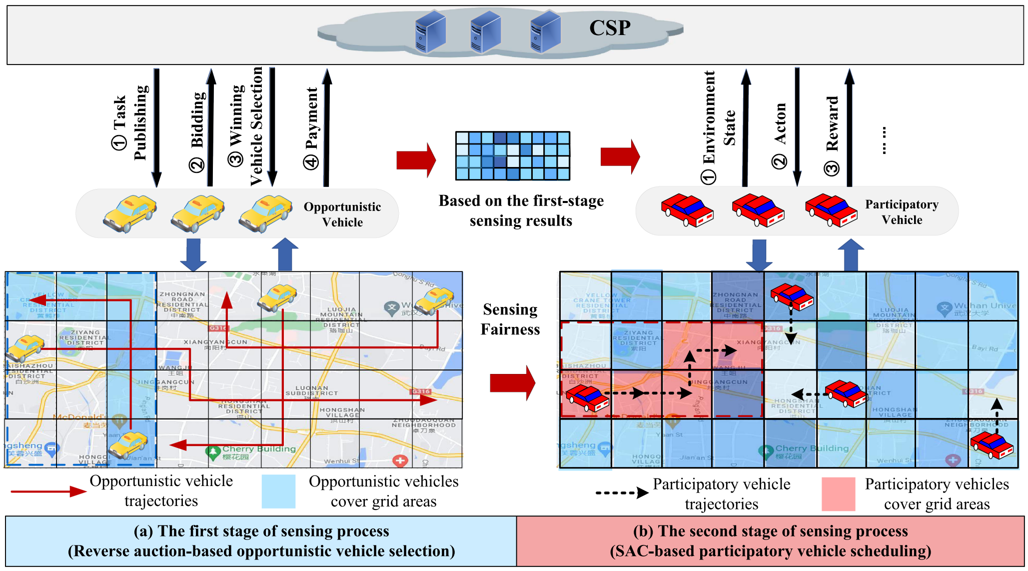


Fig. 2. The system model of two-stage hybrid sensing method.

in the impact of communication technology on both types of vehicles.

Similarity: Vehicles usually have high mobility, so the coverage and stability of communication networks will have a more significant impact. The trajectory range of both types of vehicles is large, so adequate coverage of the communication network needs to be ensured. In addition, the mobility of vehicles may cause frequent changes in network connectivity, affecting the efficiency and success of task execution. Therefore, in a time-varying vehicular crowdsensing scenario, good communication stability needs to be ensured to complete the sensing task.

Differently: Opportunistic vehicles have less stringent requirements for real-time communication. They communicate with the CSP in the initial state and upload the results after the task is completed. Therefore, it is less sensitive to latency and less demanding on network bandwidth. It only needs to upload the task results before the second-stage sensing process begins. Participatory vehicles do not have a fixed trajectory and require real-time scheduling by CSP. Therefore, they have higher requirements for the latency and bandwidth of communication networks. This requires a better communication network environment to ensure that CSP can control each participatory vehicle in real-time. Meanwhile, the sensing results need to be uploaded to the CSP in real time to update the environmental status promptly, which places high demands on network bandwidth.

B. The Model of Opportunistic Vehicle

1) Basic Definition of Opportunistic Vehicles: In the first stage of the sensing process, the opportunistic vehicle serves as the sensing entity. In this stage, the CSP will solve the opportunistic vehicle selection problem. For the opportunistic vehicle $u_m \in U$, its candidate trajectory is expressed as r_h. For simplicity, it is assumed that the vehicle can complete a sensing task if it reaches a sub-region grid. This paper defines the number of grids that candidate trajectory r_h of vehicle u_m passes through as s_m^h, which represents the number of tasks it can complete. This paper also defines the cost of vehicle u_m performing a sensing task as c_m^l, which includes various overheads such as battery, storage, and CPU. For the opportunistic vehicle u_m, if it is selected by the CSP, the cost of performing the sensing task can be expressed as:

$$C_m = c_m^l s_m^h. \tag{1}$$

Meanwhile, this paper defines a binary variable $x_m = \{0, 1\}$. This variable is used to indicate whether the CSP selects the opportunistic vehicle u_m to perform the sensing task. Specifically, $x_m = 1$ means that the opportunistic vehicle u_m is selected to perform the sensing task, whereas $x_m = 0$ means it is not selected. Among them, $x_m = 1$ means that the vehicle u_m is selected to perform the sensing task, otherwise, $x_m = 0$ is not selected.

2) Reverse Auction Model for Opportunistic Vehicles: Due to the selfish and rational characteristics of opportunistic vehicles, it is difficult to actively complete sensing tasks without incentives. Therefore, this paper introduces a reverse auction-based incentive mechanism in the first stage. As shown in Fig. 2(a), in the opportunistic vehicle selection process, CSP selects the winning vehicle based on the incentive mechanism of reverse auction and pays corresponding rewards. Specifically, the workflow is as follows.

1) First, the CSP announces the start of the auction to opportunistic vehicles set U.
2) Second, the CSP and opportunistic vehicles enter the auction stage. In the auction stage, the CSP plays the role of the auctioneer, and the opportunity vehicle plays the role of the bidder. Each opportunistic vehicle u_m reports its bid price b_m to CSP, which is the unit cost of the opportunistic vehicle u_m claiming to perform the sensing task. Please

note that the condition $b_m \geq c_m^l$ means that the bid b_m of the opportunistic vehicle u_m is not lower than its unit cost c_m^l.

3) According to the bid set of opportunistic vehicles, the CSP will determine the winning vehicle set $U' \in U$, and complete the corresponding sensing task based on the winning vehicle trajectory.
4) Finally, when the opportunistic vehicle completes the sensing task, the CSP performs payment to all winning vehicles, expressed as q_m. If the vehicle does not complete any sensing tasks, it will not receive any reward.

According to the above workflow, the CSP selects a collection of winning opportunistic vehicles through reverse auction, and then pays them corresponding rewards. Moreover, the CSP needs to control the reward paid within a reasonable budget, thereby ensuring the long-term incentive effectiveness of the system. In addition, the reverse auction-based method proposed in this paper also needs to have the following attributes:

- *Individual rationality:* For each winning opportunistic vehicle u_m, the reward paid by CSP is q_m. Each winning vehicle has non-disutility, that is, the reward q_m obtained by the opportunistic vehicle is not lower than its cost c_m^l.
- *Truthfulness:* For each winning vehicle, its truthfulness is expressed as the maximum profit that can only be obtained by bidding at the true cost. Therefore, vehicles cannot increase their own profits by maliciously raising bids.

C. The Model of Participatory Vehicle

In the second stage of the sensing process, the participatory vehicle serves as the sensing entity. In this stage, the CSP will solve the participatory vehicle scheduling problem. Here, participatory vehicles do not have a predetermined driving trajectory. Therefore, through the CSP scheduling, they can go to the target sub-region to complete the sensing task. This paper will use a DRL-based method to determine the driving route of participatory vehicles. As shown in Fig. 2(b), participatory vehicles are located in different sub-region grids. In each time slot, the CSP will determine the action of participatory vehicles v_n, that is, the sub-region grid where the next time slot should be located. Here, this paper defines the action of participatory vehicle v_n in time slot t as $a_n^t = \{1, 2, 3, 4\}$, where 1="forward", 2="backward", 3="turn left", 4="turn right". It means that CSP controls the participatory vehicle v_n to move horizontally from the current sub-region to the adjacent sub-region at time slot t.

For the participatory vehicle v_n, in addition to its fixed unit cost c_n^l of performing the sensing task, the additional distance cost generated by its detour also needs to be considered. Here, given the unit detour cost c_d, the distance cost incurred by participatory vehicles can be expressed as:

$$C_r = c_d(Dis_n - D_n^{\max}), \tag{2}$$

where Dis_n is the actual driving distance of vehicle v_n, calculated using Manhattan distance. $D_n^{\max}$ represents the planned driving trajectory distance of vehicle v_n. The CSP pays no distance cost if it is within its planned travel distance. If the planned distance is exceeded, CSP needs to pay the corresponding distance cost to the participatory vehicle v_n. In addition, the number of sensing tasks $count(v_n)$ performed by participatory vehicles v_n is limited and cannot exceed g_n counts. For the participatory vehicle v_n, the total cost of performing the sensing task can be expressed as:

$$C_n = c_n^l h_n + c_d(Dis_n - D_n^{\max}), \tag{3}$$

where $c_n^l h_n$ is the cost of the participatory vehicle performing the sensing task, c_n^l is the unit cost of performing the task, and h_n is the current number of sensing tasks completed by the vehicle. The second item is the distance cost paid to perform the sensing task, as shown in (2).

D. Problem Formulation

The CSP selects suitable opportunistic vehicles in the first stage, and then schedules a group of participatory vehicles to perform sensing tasks in the second stage. The goal is to minimize the overhead of CSP.

1) CSP Cost: First, we define P_m, as the reward by the CSP paid to opportunistic vehicle u_m, expressed as:

$$P_m = b_m s_m^h, \tag{4}$$

where b_m is the unit bid for the opportunistic vehicle u_m to perform the task, and s_m^h is the number of sensing tasks that opportunistic vehicle u_m can complete.

Since opportunistic vehicles need to participate in the reverse auction process, the rewards C_{P1} paid by the CSP to all opportunistic vehicles are:

$$C_{P1} = \sum_{u_m \in U} x_m b_m s_m^h, \tag{5}$$

where x_m determines whether the vehicle is selected as the winning vehicle by the CSP. The CSP only pays corresponding rewards to the selected winning vehicle.

Participatory vehicles are scheduled by CSP to complete the sensing task of the target sub-region. Here, according to (3), the reward C_{P2} paid by the CSP to all participatory vehicles can be obtained as:

$$C_{P2} = \sum_{v_n \in V} C_n = \sum_{v_n \in V} [c_n^l h_n + c_d(Dis_n - D_n^{\max})]. \tag{6}$$

2) The Sensing Fairness of CSP: Here, we introduce the sensing fairness index $\mathrm{Fair\_Index}(L)$ to reflect sensing fairness based on the Jain fairness index [27], [28]. According to the status of opportunistic vehicles after completing the sensing task, this paper defines $count_l^u$ as the total number of completed sensing tasks (sensing counts) of all opportunistic vehicles in sub-region grids l. Therefore, in the first stage of the sensing process, for opportunistic vehicles, the sensing fairness index is defined as:

$$\mathrm{Fair\_Index}(U_L) = \frac{(\sum_{l \in L} \mathrm{count}_l^u)^2}{|L| \sum_{l \in L} (\mathrm{count}_l^u)^2}, \tag{7}$$

where $|L|$ is the number of divided grids. It can be known that $\mathrm{Fair\_Index}(U_L) \in [0, 1]$, when $\mathrm{Fair\_Index}(U_L)$ is closer to 1, it means better fairness.

Similarly, for participatory vehicles, $count_l^v$ is defined as the total number of completed sensing tasks (sensing counts) by all participatory vehicles in sub-region grids l. Therefore, in the second stage of the sensing process, based on the sensing result of the first stage, the sensing fairness index can be defined as:

$$\mathrm{Fair\_Index}(L) = \frac{\{\sum_{l\in L}(\mathrm{count}_l^u + \mathrm{count}_l^v)\}^2}{|L|\sum_{l\in L}(\mathrm{count}_l^u + \mathrm{count}_l^v)^2}. \tag{8}$$

3) CSP Overhead: In summary, combined with the sensing fairness index, this paper redefines the overhead of CSP. Therefore, for opportunistic vehicles, the overhead of CSP in the first stage sensing process is defined as:

$$C_P(U) = \frac{C_{P1}}{\mathrm{Fair\_Index}(U_L)}. \tag{9}$$

For participatory vehicles, based on the first-stage sensing results, the overhead of CSP in the second stage sensing process is defined as:

$$C(P) = \frac{C_{P1} + C_{P2}}{\mathrm{Fair\_Index}(L)}, \tag{10}$$

where C_{P1} is the cost of the CSP payments to opportunistic vehicles in the first sensing stage, and C_{P2} is the cost of the CSP payments to participatory vehicles in the second sensing stage. $\mathrm{Fair\_Index}(L)$ is the overall sensing fairness index of the two sensing stages.

It can be seen that the overhead of CSP decreases as the fairness index $\mathrm{Fair\_Index}(L)$ increases, and decreases as the cost of the CSP payments to the vehicles decreases. Therefore, in order to reduce the overhead of CSP, in the first stage, it will select low-cost opportunistic vehicles to participate in the sensing task. In the second stage, CSP will schedule the trajectories of participatory vehicles to continue completing the sensing task. Meanwhile, the fairness of sensing coverage is ensured as much as possible.

E. Optimization Problem Definition

The optimization goal of this paper is to minimize the CSP overhead, which is achieved by reducing the cost of CSP payments to vehicles and increasing the sensing fairness among sub-regions.

First, in the first stage of sensing process, the optimization goal is defined in (11):

$$\text{P1:}\quad \min C_P(U), \tag{11}$$

$$\text{s.t.}\quad x_m \in \{0,1\}, \tag{12}$$

$$\mathrm{count}_l^u \le \eta, \tag{13}$$

$$C_{P1} \le B, \tag{14}$$

where (12) represents the binary variable of whether the opportunistic vehicle is selected. (13) represents the limit of the number of sensing times in each sub-region grid. (14) represents the limit of remuneration paid by CSP to opportunistic vehicles.

Then, in the second stage of sensing process, based on the sensing results of the first stage, the optimization goal is defined in (15):

$$\text{P2:}\quad \min C(P), \tag{15}$$

$$\text{s.t.}\quad \mathrm{count}_l^u + \mathrm{count}_l^v \le \eta, \tag{16}$$

$$C_{P1} + C_{P2} \le B, \tag{17}$$

$$\mathrm{count}(v_n) \le g_n, \tag{18}$$

where (16) represents the limit of the number of sensing times in each sub-region grid, and $count_l^u$ is known. (17) represents the limit of remuneration paid to all vehicles, and C_{P1} is known. (18) indicates that the number of sensing tasks performed by participatory vehicles v_n cannot exceed g_n.

The optimization problem proposed in this paper is an NP-Hard problem, and the special case of this optimization problem can be converted into the corresponding Minimum Weighted Set Cover (MWSC) problem to prove.

Theorem 1: The optimization problem we formulate above is NP-hard.

Proof: To prove that the problem we proposed is an NP-Hard problem, we first prove that the special case of the problem is an NP-Hard problem. We assume that the fairness index is 1, which means that all grids have equal sensing counts. Therefore, the problem can be transformed into selecting a set of bids with minimum social cost to cover all tasks (grid). Now, we introduce a well-known NP-hard problem, the Minimum Weighted Set Cover (MWSC) problem [20]. Specifically, the problem is given a set of elements $\mathcal{L} = \{l_1, \ldots, l_i, \ldots, l_k\}$, and the collection of some subsets $\Gamma = \{L^1, \ldots, L^j, \ldots, L^q\}$, where $L^j \in \mathcal{L}$ for $\forall i \in [1, q]$ and L^j has a weight c_j. It is necessary to find the minimum weight set F of the subset in Γ such that F covers all elements in $\mathcal{L}$. In our scenario, this corresponds to a given set of grids, and the trajectory of each vehicle is a corresponding grid subset. Here, each subset has a different weight, representing different task completion costs. We need to choose the lowest-cost set that will cover all grids. Obviously, the MWSC problem corresponds to a special case of the problem we raised. Therefore, the special problem we raised is an NP-Hard problem. In addition, the general case of the proposed optimization problem is an NP-Hard problem, thus proving that the theorem holds. ∎

IV. Opportunistic Vehicle Selection

In the first stage of the sensing process, for the opportunistic vehicle selection problem, this section proposes a reverse auction-based incentive mechanism. It includes two steps: winning vehicle selection and reward payment. In addition, this section also proves that this method ensures the individual rationality and truthfulness of opportunistic vehicles. Fig. 3 shows the technical flow of this section.

A. Winning Opportunistic Vehicle Selection

In the first stage of the sensing process, a reverse auction-based incentive mechanism is designed. This allows the CSP to select the lowest-cost opportunistic vehicles to complete sensing tasks. This stage is mainly divided into two steps. In the first

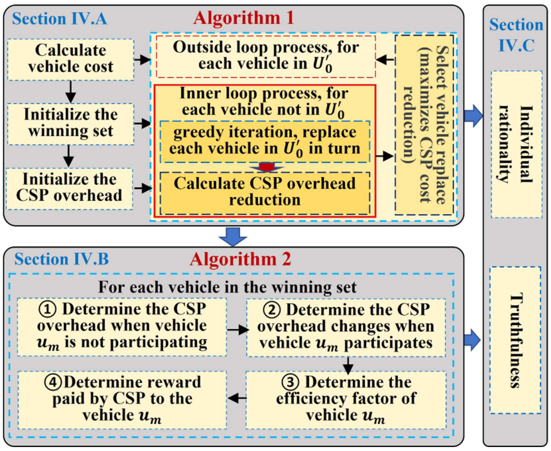


Fig. 3. The technical flow in Section IV.

step, the CSP needs to determine the optimal set of opportunistic vehicles as the winning vehicle set. In the second step, the CSP needs to determine the appropriate payment mechanism to pay the winning vehicle. This stage aims to optimize problem P1, as shown in (11) to (14).

1) Algorithm Overview: Specifically, in the reverse auction process, the opportunistic vehicle acts as the buyer, and the CSP acts as the seller. Since selecting the winning vehicle to perform the sensing task is NP-Hard, this paper proposes an approximate method called the Opportunistic Vehicle Winner Selection Algorithm (OVWSA). This method mainly relies on greedy thinking to select the set U' of $|M'|$ vehicles that minimize CSP overhead. Here, U' represents the optimal winning vehicle set, and $|M'|$ represents the number of winning vehicles in this set. The specific process is as follows.

2) Initialize the Winning Set and Calculate the Initial CSP Overhead: First, the winning set of opportunistic vehicles is initialized. The task completion cost C_m of all vehicles in the candidate opportunistic vehicle set is calculated through (1) (lines 1-3). Next, the C_m are sorted in increasing order, and the new candidate vehicle set is represented as $U^N = \{u_1, u_2, \ldots, u_m, \ldots, u_{M'}, u_{M'+1}, u_{m'}, \ldots, u_M\}$ (line 4). Then, in the new candidate vehicle set U^N, select the initial $|M'|$ vehicles as the initial winning vehicle set $U_0' = \{u_1, u_2, \ldots, u_m, \ldots, u_{M'}\}$, and calculate the initial CSP overhead $C_P(U_0)$ (line 5).

3) Greedy Iteration, Inner Loop Process: Next, the algorithm enters the iterative update process, which contains two levels of loop processes. In the outer loop, starting from vehicle $u_{M'+1}$ to vehicle u_M, i.e., for each vehicle in $\{u_{M'+1}, \ldots, u_m, \ldots, u_M\}$, the inner loop proceeds as follows (line 8–10). The algorithm replaces each vehicle in the set U_0' in sequence from back to front, that is, $\{u_{M'}, \ldots, u_m, \ldots, u_1\}$. Specifically, in the inner loop, taking vehicle $u_{M'+1}$ as an example, vehicle $u_{M'+1}$ is used to replace each vehicle in set U' from back to front. Then, based on the bid price of each vehicle, after the vehicle $u_{M'+1}$ replaces the corresponding vehicle u_m, the corresponding CSP overhead $C_P(U_{m'})$ is calculated by (9).

At this time, we define $\Delta U_{m'}$ given in (19):

$$\Delta U_{m'} = C_P(U_{m'}) - C_P(U_0), \tag{19}$$

which represents the change in CSP overhead after replacing vehicle u_m in the initial winning vehicle set U_0 with vehicle $u_{M'+1}$. For vehicle $u_{M'+1}$, when it replaces the vehicles in the set $\{u_{M'}, \ldots, u_m, \ldots, u_1\}$ for one round, the set $\Delta \mathbf{U}(M) = \{\Delta U_1, \ldots, \Delta U_{M'}\}$ is generated. If there is any $\Delta U < 0$, it means that the newly added vehicle reduces the CSP overhead compared to the original vehicle. At this time, vehicle $u_{M'+1}$ is used to replace the vehicle with the smallest corresponding value in the set $\Delta U(M)$ (line 11-13). The above process is a round inner loop process.

4) Outside Loop Process: Then, the above inner loop process is repeated for vehicles $u_{M'+2}$ to u_M. After the inner loop ends, the $|M'|$ vehicles finally retained in the set U' are the winning vehicle set, and these vehicles can minimize the CSP overhead.

Note that in the above process, the completion count and budget for sensing tasks in sub-region grids are limited. If the limit is exceeded, the iteration will be stopped and the current vehicle will be deleted or the process will be terminated. The specific algorithm steps are shown in Algorithm 1. In addition, OVWSA contains a two-layer loop process and iterates for each vehicle, so the time complexity of the algorithm is $O(N^2)$.

B. Reward Payment

After the winning vehicle selection stage, the CSP needs to pay the corresponding remuneration for the winning opportunistic vehicle. Opportunistic vehicles will consume corresponding costs when completing sensing tasks, and the cost of completing the tasks can be calculated. Given the selfish and rational characteristics of opportunistic vehicles, it is generally assumed that they hope to obtain higher returns. Therefore, unified payment rules must be formulated to ensure the rationality of payment. Here, referring to the standard Vickry-Clarke-Groves (VCG) scheme, this paper designs corresponding payment rules [19]. Specifically, in this subsection, the Opportunistic Vehicle Reward Payment Algorithm (OVRPA) is proposed. This approach can ensure the individual rationality and truthfulness of opportunistic vehicles. The specific process is shown below.

1) Determine the CSP Overhead When Vehicle u_m is not Participating: For the winning vehicle set U', we first define $C_P^{-m}(U)$ to represent the minimum cost of CSP without the participation of opportunistic vehicles u_m. Here, based on Algorithm 1, the optimal winning vehicle set without the participation of opportunistic vehicles u_m can be obtained (line 2). Therefore, the optimal cost $C_P^{-m}(U)$ of CSP without considering the participation of opportunistic vehicles u_m is calculated as (line 3):

$$C_P^{-m}(U) = \frac{C_{P1}^{-m}}{\text{Fair\_Index}(U_L)^{-m}}. \tag{20}$$

2) Determine the CSP Overhead Changes When Vehicle u_m Participates: Then, when the vehicle u_m participates in the sensing process, the CSP overhead is $C_P(U)$, and the following

Algorithm 1: Opportunistic Vehicle Winner Selection Algorithm **(OVWSA)**.

Input: Opportunistic vehicle candidate set U_M, budget B, task execution cost c_l of vehicle u_m, the number of sensing tasks s_m^h that can be completed by the candidate trajectory of vehicle u_m;
Output: Winning opportunistic vehicle set U';
1: **For each** $u_m \in U_M$ **do:**
2: Calculate the cost C_m of each vehicle to complete the sensing task according to (1).
3: **End for**
4: The candidate vehicles are sorted from small to large according to the cost C_m, and the initial candidate vehicle set $U^N = \{u_1, u_2, \ldots, u_{M'}, u_{M'+1}, \ldots, u_M\}$ is obtained.
5: In the new candidate set, select the top $|M|$ vehicles as the initial winner set U_0', and calculate the initial CSP cost $C_P(U_0)$ according to (9).
6: **For each** $u_{m'} \in \{u_{M'+1}, \ldots, u_M\}$ **do:**
7: **For each** $u_m \in \{u_{M'}, \ldots, u_1\}$ **do:**
8: Substitute the first $|u_{M'}|$ vehicles in the initialised candidate opportunistic vehicle set U^N in turn, i.e., $\{u_{M'}, u_m, \ldots, u_1\}$.
9: After replacing u_m with $u_{m'}$, calculate the corresponding $C_P(U_m)$, and then obtain $\Delta U_{m'} = C_P(U_{m'}) - C_P(U_0)$.
10: Get the set $\Delta \mathbf{U}(M) = \{\Delta U_1, \ldots, \Delta U_{M'}\}$.
11: **If** $\min\ \Delta U(M) < 0$ **then:**
12: $u_m \leftarrow \arg\min_{u_{m'}} \Delta U(M)$
13: **End If**
14: **End for**
15: **End for**
16: **return** U'

equation can be obtained:

$$p_m = C_P^{-m}(U) - C_P(U) > 0, \tag{21}$$

where p_m represents the degree of CSP overhead reduction when vehicle u_m participates in the auction process (line 4).

3) Determine the Efficiency Factor of Vehicle u_m: Currently, the bid b_m for the opportunistic vehicle u_m and the corresponding number of completed sensing tasks s_m^h can be obtained. Then, the efficiency factor ρ_m of the opportunistic vehicle u_m can be expressed as:

$$\rho_m = \frac{p_m}{b_m s_h^m}, \tag{22}$$

where the numerator represents the cost reduction brought by the opportunistic vehicle u_m to the CSP, and the denominator represents the reward it requires from the CSP (line 5).

4) Determine Reward Paid by CSP to the Vehicle u_m: Therefore, it can be determined that the compensation paid by CSP to the winning vehicle u_m is expressed as (line 6):

$$q_m = (1 + \rho_m) b_m s_h^m. \tag{23}$$

Algorithm 2: Opportunistic Vehicle Reward Payment Algorithm **(OVRPA)**.

Input: Opportunistic vehicle candidate set U_M, opportunistic vehicle winner set U', budget B, vehicle u_m's task execution cost c_l and quotation b_m, the number of sensing tasks that can be completed by vehicle u_m's trajectory s_m^h;
Output: The reward set $Q(m)$ paid to the winning vehicles;
1: **For each** $u_m \in U'$ **do:**
2: Calculate the winning vehicle set U^{-m} of CSP without the participation of vehicle u_m according to Algorithm 1.
3: Calculate the optimal CSP overhead $C_P^{-m}(U)$ without the participation of vehicle u_m according to (20).
4: Calculate p_m according to (21).
5: Calculate ρ_m according to (22).
6: Calculate q_m according to (23).
7: $B^* \leftarrow B - q_m$.
8: **If** $B^* < 0$ **then:**
9: **Break**
10: **End for**
11: **return** $Q_{(m)} = \{q_1, \ldots, q_m\}$

It can be seen that for the winning vehicle u_m, the CSP will pay a reward greater than its cost, thereby increasing the enthusiasm of opportunistic vehicles to participate in the sensing task. For vehicles that are not selected, no compensation will be paid to them. The specific algorithm process is shown in Algorithm 2.

C. Theoretic Analysis

This subsection proves that the proposed reverse auction-based opportunistic vehicle selection method needs to satisfy both the following important attributes: individual rationality and truthfulness. As the main motivation for vehicles to help CSP complete sensing tasks, the individual rationality attribute needs to ensure that each opportunistic vehicle can obtain non-negative rewards. In addition, the truthfulness attribute needs to ensure that each opportunistic vehicle cannot obtain higher rewards from untrue bids.

Theorem 2. The proposed reverse auction-based opportunistic vehicle selection method is consistent with truthfulness. That is, each opportunistic vehicle needs to set its bid to its cost ($b_m = c_m^l$).

Proof. First, this paper assumes that an opportunistic vehicle u_m in the winning vehicle set reports an unreal bid $b_m'(b_m' > c_m^l)$. Then, according to (1), we can get the current unreal cost C_n', and there exists $C_n' > C_n$. Further, we can get $C_{P1}' > C_{P1}$ according to (5). By calculating the current CSP cost $C_P'(U)$, when vehicle u_m reports an unreal bid, the CSP cost difference can be expressed as:

$$\hat{C}_P(U) = C_P'(U) - C_P(U),$$

$$= \frac{C'_{P1} - C_{P1}}{\text{Fair\_Index}(U_L)} > 0, \tag{24}$$

it can be seen that the unreal bid of vehicle u_m will increase the CSP overhead.

According to (21), it can be concluded that the vehicle has $p'_m - p_m < 0$ in the case of unreal bidding. Then, we get $\rho'_m < \rho_m$ according to (22). Finally, according to (23), we can get:

$$\hat{q}_m = q'_m - q_m = [1 + (\rho'_m - \rho_m)]c_m < 0. \tag{25}$$

In summary, it can be known that vehicle u_m will not increase its own compensation through unreal bidding. Therefore, it shows that vehicle u_m cannot increase its own income through unreal bidding prices, ensuring truthfulness. ■

Theorem 3: The payment rule defined in (23) is consistent with individual rationality, that is, $\forall u_m \in U', q_m - c_m \geq 0$.

Proof: According to the payment rules proposed by our OVRPA, for any vehicle u_m in the winner set, (21) is always true, this means that $\rho_m > 0$ in (22) is always true. Therefore, according to (23), it can be concluded that $q_m > c_m$ is also always true. This shows that each winning vehicle will receive a compensation payment greater than its own cost, so individual rationality is satisfied. ■

V. Participatory Vehicle Scheduling

In the second stage of the sensing process, a DRL-based method is proposed to solve the participatory vehicle scheduling problem. This section first approximates the optimization problem as Markov Decision Processes (MDP) to minimize CSP overhead. Then, the state, action, and reward functions in MDP are defined. Finally, a SAC-based method named SPVSA is proposed to solve this problem.

A. State, Action, and Reward Function Definitions

In the first stage, CSP recruited a group of opportunistic vehicles to complete the sensing tasks. Based on the first stage, the second stage continues to schedule participatory vehicles to complete the sensing task and ensure sensing fairness. Here, using the DRL method, the CSP acts as a decision-maker and schedules the trajectories of n participatory vehicles to complete the sensing task. The goal of this stage is to optimize problem P2, as shown in (15) to (18). Note that this stage divides the entire sensing window into t time slots.

When participatory vehicles perform sensing tasks, due to the uncertainty of their driving trajectories, CSP can be used as an intelligent agent to make unified decisions and control all participatory vehicles. In the second stage of the sensing process, it is necessary to make immediate action decisions for the vehicle. Therefore, this paper discretizes the time window $[\frac{1}{2}D, D]$ into T time slots and expresses it as $\{1, \ldots, t, \ldots, T\}$. Here, we will introduce the relevant state space, action space, and reward function of the DRL-based participatory vehicle scheduling problem.

1) *State Space (S):* The status is used to represent the network environment status faced by the current time slot. Correspondingly, the status of time slot t is represented by $\mathbf{s_t} = \{\mathbf{S_1}, \mathbf{S_2}, B_r^t\}$. Among them, $\mathbf{S_1} = \{loc_{v_1}^t, \ldots, loc_{v_n}^t\}$ represents the position index of each participatory vehicle v_n in time slot t (that is, the position index of the sub-region grid). $\mathbf{S_2} = \{count_1^t, \ldots, count_l^t\}$ represents the number of sensing times of each sub-region grid at time slot t. B_r^t represents the remaining budget of CSP at the beginning of time slot t.
2) *Action Space (A):* The CSP is responsible for determining the action strategy of each participatory vehicle v_n. That is, it is responsible for scheduling participatory vehicle trajectories to allocate corresponding sensing tasks. The action of time slot t is represented by $\mathbf{a_t} = \{a_1^t, \ldots, a_n^t, \ldots, a_N^t\}$. Here, $a_n^t = \{1, 2, 3, 4\}$, which indicates that CSP controls the participatory vehicle v_n to move horizontally from the current sub-region to the adjacent sub-region at time slot t (1="forward", 2= "Back", 3 = "Turn left", 4 = "Turn right").
3) *Reward Function (R):* The goal of this article is to minimize CSP overhead, which is opposite to the goal of maximizing cumulative discount rewards for DRL. According to (7) and (10), it can be known that when the first-stage sensing result is determined, the cost of participatory vehicles and the sensing fairness index affect the CSP overhead. For simplicity, we assume that participatory vehicles have the same task completion cost. At this time, as time goes by, the sensing task completion cost of participatory vehicles increases linearly, which has no impact on the optimization goal. Therefore, we set the reward function to maximize the sensing fairness index:
$$r_t = \text{Fair\_Index}(L), \tag{26}$$
where the sensing fairness index is between [0,1]. When r_t is closer to 1, it indicates that the sensing state is fairer. Correspondingly, the overhead of CSP is also smaller. Meanwhile, this setting will also avoid the problem of large fluctuations in results during the training process.

B. From RL to DRL

MDP can be used to describe many decision-making control problems, which can usually be solved using dynamic programming or linear programming methods. Nevertheless, in scenarios involving dynamic and uncertain states and decisions, RL methods prove more effective. RL empowers the agent to actively learn within an environment, aiming to optimize its actions for maximal cumulative rewards.

Specifically, Q-learning is a typical RL method, which can be described as the learning behavior of obtaining the optimal policy in MDP. Using a Q-table for value iteration, the agent calculates Q-function results for each state-action pair. Post interaction with the environment, the agent updates and maintains Q-values in the table to refine its learning process.

Q-learning is constrained by the need to manage Q-tables, limiting its applicability to problems with small-scale action and state spaces. In contrast, DRL excels in automatically extracting features from high-dimensional data, making it well-suited for

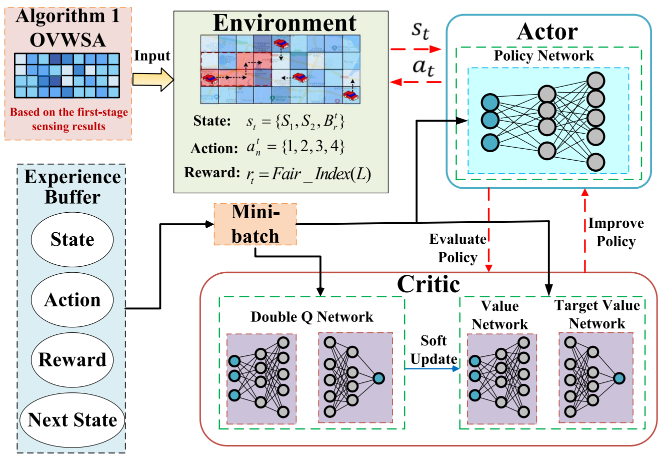


Fig. 4. The structure of SPVSA, which is based on the results of the first stage OVWSA. The CSP serves as an agent to schedule participatory vehicles. It determines the action a_t according to the state s_t to obtain the optimal reward r_t.

large-scale state space problems. In summary, this paper proposes a SAC-based method to solve the participatory vehicle scheduling problem.

C. SAC-Based Participatory Vehicle Scheduling Method

Traditional DQN is usually used to deal with problems with a single discrete action space, and can only output one discrete action at each time step. Our problem targets multiple vehicles and involves multiple discrete actions, so traditional DQN methods for discrete action spaces encounter challenges. Currently, the SAC algorithm has been widely used to solve sequence decision-making problems in other related scenarios such as vehicular edge computing [29], [30], [31]. We introduced it into the vehicle crowd sensing scenario for the first time. In comparison to other DRL algorithms such as DQN and DDPG, SAC incorporates an entropy regularization term into its objective function to enhance exploratory capabilities. Since entropy reflects the randomness of the policy, policies characterized by enhanced entropy provide agents with more opportunities. This enables the agent to perform high-reward actions in the most random manner possible, while enhancing the agent's exploration capabilities. However, the SAC algorithm is usually applied to continuous action spaces and cannot be directly applied to the discrete action spaces of this paper. Therefore, in order for the SAC algorithm to match our problem, the discretization problem of the action space is solved through the discrete action continuation method. This method can avoid the problem that the action space is too large due to the increase in the number of vehicles, making convergence difficult. Based on this, this paper proposes a SAC-based Participatory Vehicle Scheduling Algorithm (SPVSA). Fig. 4 shows the network structure of SPVSA.

Our goal is to schedule the trajectory of each vehicle to maximize the expected reward value and its entropy value. Correspondingly, the objective function contains two parts:

$$J(\pi) = \sum_{t=0}^{T} \mathbb{E}_{(\boldsymbol{s}_t, \boldsymbol{a}_t) \sim \rho_\pi} \left[r(\boldsymbol{s}_t, \boldsymbol{a}_t) + \zeta \mathcal{H}(\pi(\cdot \mid \boldsymbol{s}_t)) \right], \quad (27)$$

where ρ_π signifies the state and state-action distribution resulting from the policy π, with $r(\boldsymbol{s}_t, \boldsymbol{a}_t)$ representing the reward term. The parameter ζ functions as the temperature parameter, offering a means to calibrate the significance of the entropy term in relation to the reward. $\mathcal{H}(\pi(\cdot \mid \boldsymbol{s}_t)) = -\log \pi(\boldsymbol{a}_t|\boldsymbol{s}_t)$ is the entropy term.

Consequently, the entropy-augmented reward function $J(\pi)$ is formulated as:

$$J(\pi) = \sum_{t=0}^{T} \mathbb{E}[r(\boldsymbol{s}_t, \boldsymbol{a}_t) - \zeta \log \pi(\boldsymbol{a}_t|\boldsymbol{s}_t)|\pi]. \quad (28)$$

During the policy evaluation phase within the soft iteration, when provided with an initial state $\boldsymbol{s}$ and action $\boldsymbol{a}$, the soft Q-value function is derived in accordance with the entropy objective outlined in (28):

$$Q(\boldsymbol{s}, \boldsymbol{a}) = \mathbb{E}\left[\sum_{t=1}^{T} \gamma^t \left[r_t - \zeta \log \pi(\boldsymbol{a}_t \mid \boldsymbol{s}_t)\right] \middle| \boldsymbol{s}_0 = \boldsymbol{s}, \boldsymbol{a}_0 = \boldsymbol{a}, \pi\right], \quad (29)$$

where r_t represents the system reward. Next, according to the Bellman equation, the soft value function is generated:

$$V(\boldsymbol{s}_t) = \mathbb{E}_{\boldsymbol{a}_t \sim \pi}[Q(\boldsymbol{s}_t, \boldsymbol{a}_t) - \zeta \log \pi(\boldsymbol{a}_t|\boldsymbol{s}_t)]. \quad (30)$$

Based on the actor-critic algorithm, SAC consists of two parts: a policy improvement model (i.e., actors) and a strategy evaluation model (i.e., critics). Specifically, based on the current observed state and actions, the critic model evaluates the state of the environment by updating soft Q-values and soft state value functions. The Actor model is committed to making optimal decisions for the Agent based on its own observation states.

1) *Critic Model:* In the critic model, there are 4 networks, including two Q-networks ψ_1, ψ_2 for estimating the values of state-action pairs, the value network ϕ for estimating state values, and the target value network $\bar{\phi}$ that represents an exponential moving average of the value network ϕ. Next, a soft-valued function needs to be trained to minimize the squared residual:

$$J_V(\phi) = \mathrm{E}_{s_t \sim \mathcal{B}} \left[\frac{1}{2} \left(V_\phi(s_t) - \mathbb{E}_{a_t \sim \pi_\theta} \left[Q_\psi(s_t, a_t) - \zeta \log \pi_\theta(a_t \mid s_t) \right] \right)^2 \right], \quad (31)$$

where $\mathcal{B}$ is a fixed-size replay buffer that stores previous experience tuples $(\boldsymbol{s}_t, \boldsymbol{a}_t, r_t, \boldsymbol{s}_{t+1})$. An unbiased estimator can be used to estimate the gradient of (31):

$$\hat{\nabla}_\phi J_V(\phi) = \nabla_\phi V_\phi(\boldsymbol{s}_t) \times \left(V_\phi(\boldsymbol{s}_t) - Q_\psi(\boldsymbol{s}_t, \boldsymbol{a}_t) + \zeta \log \pi_\theta(\boldsymbol{a}_t \mid \boldsymbol{s}_t) \right). \quad (32)$$

Correspondingly, the parameters of the value network ϕ can be updated as follows:

$$\phi \leftarrow \phi - \lambda_V \hat{\nabla}_\phi J_V(\phi), \quad (33)$$

where $\lambda_V \geq 0$ is the learning rate of the value network. The parameters of the target value network $\bar{\phi}$ can be updated by:

$$\bar{\phi} \leftarrow \tau\phi + (1-\tau)\bar{\phi}, \tag{34}$$

where $\tau \in [0,1]$ is the weight used for the update of the target value network.

The soft Q-function parameters are trained to minimize the soft Bellman residual:

$$J_Q(\psi) = \mathbb{E}_{(\boldsymbol{s}_t,\boldsymbol{a}_t)\sim\mathcal{B}}\left[\frac{1}{2}\left(Q_\psi(\boldsymbol{s}_t,\boldsymbol{a}_t) - \widehat{Q}(\boldsymbol{s}_t,\boldsymbol{a}_t)\right)^2\right], \tag{35}$$

with

$$\widehat{Q}(\boldsymbol{s}_t,\boldsymbol{a}_t) = r(\boldsymbol{s}_t,\boldsymbol{a}_t) + \gamma\mathbb{E}_{\boldsymbol{s}_{t+1}\sim p}\left[V_{\bar{\phi}}(\boldsymbol{s}_{t+1})\right], \tag{36}$$

which is also used stochastic gradients for optimization:

$$\hat{\nabla}_\psi J_Q(\psi) = \nabla_\psi Q_\psi(\boldsymbol{a}_t,\boldsymbol{s}_t) \times \left(Q_\psi(\boldsymbol{s}_t,\boldsymbol{a}_t) - r(\boldsymbol{s}_t,\boldsymbol{a}_t) - \gamma V_{\bar{\phi}}(\boldsymbol{s}_{t+1})\right). \tag{37}$$

Correspondingly, the parameters of the Q network ψ_i ($i \in \{1,2\}$) can be updated as:

$$\psi_i \geq \psi_i - \lambda_Q \hat{\nabla}_\psi J_Q(\psi), \tag{38}$$

where $\lambda_Q \geq 0$ is the learning rate of the Q-network.

2) *Actor Model:* In the actor model, the policy network produces both the mean and standard deviation parameters for an action distribution. For every state, the policy undergoes an update to conform to the exponential distribution of the updated soft Q-function. This adjustment is achieved by minimizing the anticipated Kullback–Leibler (KL) divergence, a strategy demonstrated to result in an enhanced policy. The new policy can be computed as follows:

$$\pi_{\text{new}} = \arg\min_{\pi_\theta\in\Pi} D_{KL}\left(\pi_\theta(\cdot\mid s_t)\,\middle\|\,\frac{\exp\left(Q^{\pi_{\text{old}}}(\boldsymbol{s}_t,\cdot)/\zeta\right)}{Z^{\pi_{\text{old}}}(\boldsymbol{s}_t)}\right), \tag{39}$$

where Π represents the feasible sets of the policy function, and $D_{KL}(\cdot)$ stands for the Kullback–Leibler (KL) divergence. $Z^{\pi_{old}}(\boldsymbol{s}_t)$ represents the partition function that normalizes the distribution, contributing no gradient to the new policy.

Correspondingly, a DNN with parameters θ is adopted to approximate the policy function $\pi_\theta(\cdot|\boldsymbol{s}_t)$ to obtain an improved policy. Then, the policy parameters are learned by minimizing the expected KL divergence in (39) as follows:

$$J_\pi(\theta) = \mathbb{E}_{\boldsymbol{s}_t\sim\mathcal{B}}\left[\mathbb{E}_{\boldsymbol{a}_t\sim\pi_\theta}\left[\zeta\log\left(\pi_\theta(\boldsymbol{a}_t\mid\boldsymbol{s}_t)\right) - Q_\psi(\boldsymbol{s}_t,\boldsymbol{a}_t)\right]\right]. \tag{40}$$

Subsequently, we reparameterize the policy through a neural network transformation:

$$\boldsymbol{a}_t = f_\theta(\epsilon_t;\boldsymbol{s}_t), \tag{41}$$

where ϵ_t represents an input noise vector. The output of the policy function $f_\theta(\cdot)$ comprises both the mean f_θ^μ and the standard deviation f_θ^σ of the action. Therefore, the (41) can be rewritten as:

$$\boldsymbol{a}_t = f_\theta^\mu(\boldsymbol{s}_t) + \epsilon_t \odot f_\theta^\sigma(\boldsymbol{s}_t), \tag{42}$$

Algorithm 3: SAC-based Participatory Vehicle Scheduling Algorithm **(SPVSA)**.

Input: State space $\boldsymbol{s}_t$, action space $\boldsymbol{a}_t$, and SAC network parameters;
Output: Optimal vehicle scheduling action;
1: Initialize $\theta,\psi,\phi,\bar{\phi}$
2: Initialize experience replay buffer $\mathcal{B}$
3: **For each** episode **do:**
4: Initialize the environment and observe the initial state $\boldsymbol{s}_0$;
5: **For each** timeslot $t \in \{1,2,\ldots,T\}$ **do:**
6: Observe the state $\boldsymbol{s}_t$, take action $\boldsymbol{a}_t \sim \pi_\theta(\cdot|\boldsymbol{s}_t)$;
7: Obtain new status $\boldsymbol{s}_{t+1}$ and reward r_t;
8: Store experience $(\boldsymbol{s}_t,\boldsymbol{a}_t,rt,\boldsymbol{s}_{t+1})$ into relay buffer $\mathcal{B}$;
9: Takes random batch samples from relay buffer $\mathcal{B}$;
10: Update parameters of the value network defined in (33);
11: Update parameters of the two soft Q networks defined in (38);
12: Update parameters of the policy network defined in (45);
13: Update parameters of the target value network defined in (34);
14: **End for**
15: **End for**

where $\odot$ represents the element-wise Hadamard product.

Then, (40) can be rewritten as:

$$J_\pi(\theta) = \mathbb{E}_{\boldsymbol{s}_t\sim\mathcal{B},\epsilon_t\sim\mathcal{N}}\left[\log\pi_\theta\left(f_\theta(\epsilon_t;\boldsymbol{s}_t)\mid\boldsymbol{s}_t\right) - Q_\psi\left(\boldsymbol{s}_t, f_\theta(\epsilon_t;\boldsymbol{s}_t)\right)\right], \tag{43}$$

where π_θ is implicitly defined in terms of f_θ. It is further observed that the partition function is independent of θ and can therefore be omitted. Subsequently, the gradient of (43) can be approximated as:

$$\hat{\nabla}_\pi J_\pi(\theta) = \nabla_\theta\log\pi_\theta(\boldsymbol{a}_t|\boldsymbol{s}_t) + \left(\nabla_{\boldsymbol{a}_t}\log\pi_\theta(\boldsymbol{a}_t|\boldsymbol{s}_t) - \nabla_{\boldsymbol{a}_t}Q(\boldsymbol{s}_t,\boldsymbol{a}_t)\right)\nabla_\theta f_\theta(\epsilon_t;\boldsymbol{s}_t). \tag{44}$$

Then, the parameters of the policy network are updated by minimizing $J_\pi(\theta)$:

$$\theta \geq \theta - \lambda_\pi\hat{\nabla}J_\pi(\theta), \tag{45}$$

where $\lambda_\pi \geq 0$ is the learning rate of the policy network.

In addition, SAC is an off-policy algorithm using experience replay technology. Specifically, during the parameter update phase, parameter updates are performed by uniformly and randomly sampling small batches of data in the replay buffer.

Next, the detailed process of the SPVSA is introduced. First, we need to initialize the DNN parameters $\theta,\psi,\phi,\bar{\phi}$, and the experience replay buffer $\mathcal{B}$ (line 1-2). Then, in the exploration phase, the CSP receives the initial state $\boldsymbol{s}_0$ at the beginning of each episode (line 4). In each time slot t, the CSP observes the state $\boldsymbol{s}_t$ and selects an action $\boldsymbol{a}_t$ based on its actor network (line 6). Then, CSP obtains a new state $\boldsymbol{s}_{t+1}$ and receives an instant reward r_t (line 7). At this time, CSP stores the tuple $(\boldsymbol{s}_t, \boldsymbol{a}_t, rt, \boldsymbol{s}_{t+1})$ into the experience playback buffer $\mathscr{B}$ (line 8). Afterward, actor and critic network parameters are updated based on mini-batch training (line 9). Specifically, according to (33), (38), (45), (34), the parameters of the value network, the two soft Q networks, the policy network, and the target value network are updated respectively (line 10-13).

VI. Experiment Analysis

This section performs a performance analysis of the proposed two-stage hybrid sensing method. The effectiveness of the proposed method is verified through real data sets and is better than the baseline solution.

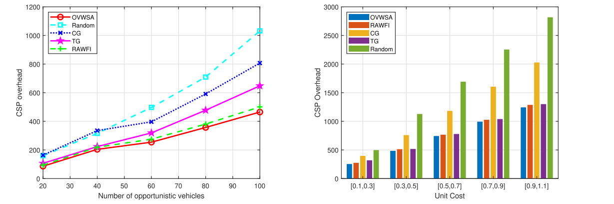
A. Experimental Setup

1) The First Stage Experimental Setup: In the first-stage sensing process, for the reverse auction-based opportunistic vehicle selection problem, this paper uses the real-world Roma data set to verify the effectiveness of the proposed method [13], [32]. This data set collects the trajectory data of 320 taxis in Rome, Italy within 30 days. Specifically, the data time is from February 1, 2014 to March 2, 2014. The vehicle trajectory is updated every about 15 seconds by latitude and longitude. The data format is: <taxi ID, date, time, latitude, longitude>.

Due to the large geographical scope of the data set, this paper uses a part of the original data set to ensure the accurate discovery of vehicle motion information. First, we select a data set whose GPS range is within the rectangular region between (41.87, 12.44) and (41.93, 12.50) [7]. Then, the rectangular region is divided into sub-region grids of the same size, and each sub-region grid has a sensing task. Depending on the division longitude, the sparsity of the task is also different. Here, we set the grid division accuracy interval to [0.001, 0.01]. When the vehicle passes through a certain sub-region grid once, it represents the completion of a sensing task. Finally, the vehicle trajectory is generated based on the taxi's location and time. It is then matched with the sub-region grid to determine the sensing tasks that each vehicle can complete.

We randomly select the trajectories formed by the first 1000 trajectory points of [20,100] vehicles in the data set for analysis. Then, the number of winning vehicles is set to half of the number of candidate vehicles. In addition, the unit cost of each opportunistic vehicle completing the task varies between [0.3, 1.1]. In order to compare with the proposed reverse auction-based OVWSA, we propose the following benchmark scheme:

1) Reverse Auction Without Fairness Index (RAWFI) [16]: CSP still selects opportunistic vehicles through the reverse auction method, but does not consider the impact of sensing fairness.
2) Cost-based Greedy method (CG) [33]: CSP only selects the opportunistic vehicle with the lowest unit cost to complete the sensing task, that is, the vehicle with the lowest bid.



(a) The number of candidate opportunistic vehicles. (b) The unit cost of performing sensing tasks.

Fig. 5. The CSP overhead changes under different opportunistic vehicle scenarios.

3) Task-based Greedy method (TG): CSP only selects the opportunistic vehicle that can complete the largest number of sensing tasks.
4) Random: CSP randomly selects a target number of opportunistic vehicles to complete the sensing task.

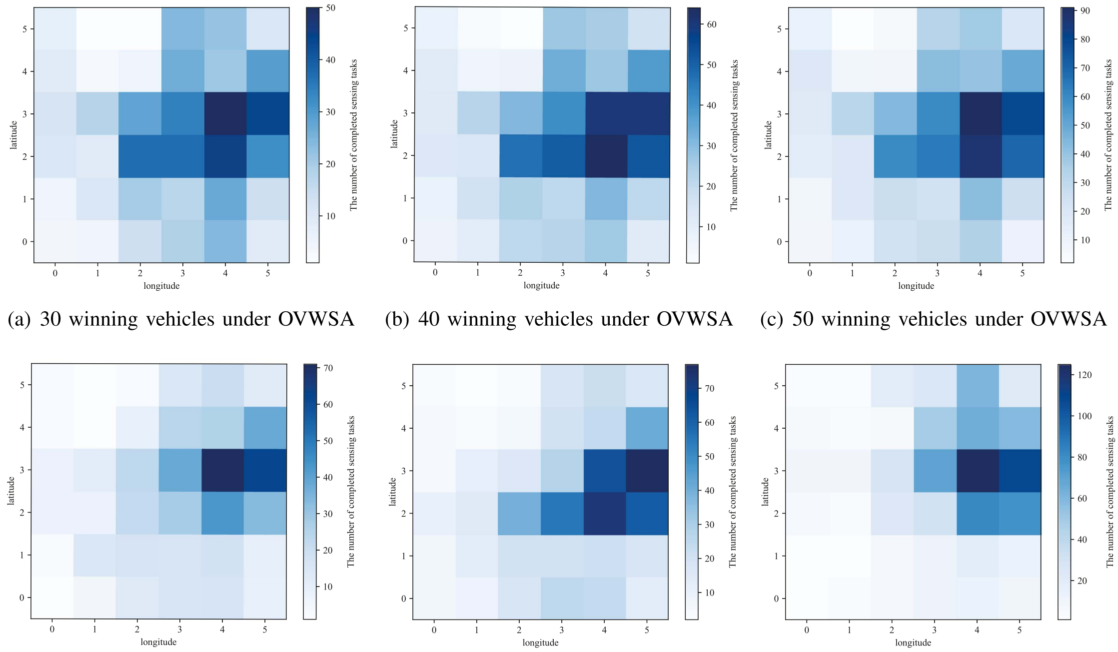
2) The Second Stage Experimental Setup: For the SAC-based participatory vehicle scheduling problem, this paper sets the same scenario as the opportunistic vehicle, that is, analyzes within the rectangular region of the same GPS range. Then, the rectangular target region is divided. To match the first-stage sensing process, we consider a $6 * 6$ grid with an accuracy of 0.01, and a $12 * 12$ grid with an accuracy of 0.005. The number of sensing tasks that each participatory vehicle can complete is set to 50. Furthermore, we set the discount factor $\gamma = 0.99$, the learning rate 1×10^{-4}, and the batch size 256.

In order to compare with the proposed SPVSA, this paper uses the following benchmark scheme:

1) Actor-Critic (AC) [34]: AC has engineered a comprehensive agent that can perform policy output and real-time policy evaluation through value functions. Specifically, the Actor leverages the strategy function to produce actions, engaging with the environment. The Critic employs the value function to assess the Actor's efficacy, thereby influencing subsequent stage actions undertaken by the Actor.
2) Greedy [10]: For each participatory vehicle, the CSP makes actions in a greedy manner. It will select the grid of adjacent sub-regions that has the lowest number of completions of the sensing task.
3) Random: CSP randomly makes action decisions, determines the trajectory of participatory vehicles, and completes corresponding sensing tasks.

B. Analysis of Opportunistic Vehicle Experiment Results

This section analyzes the performance of the OVWSA proposed in this paper. Fig. 5(a) compares the changes in CSP overhead when the number of candidate vehicles changes. Here, the unit task cost is set between [0.1, 0.3], and the sub-region grid division accuracy is 0.005. It can be seen that as the number of vehicles increases, the overhead of CSP gradually increases. This is because there are more opportunistic vehicles participatory in sensing tasks, and CSP must pay more remuneration, resulting in an increase in overhead. In addition, the proposed OVWSA has the lowest overhead, which verifies the optimality



(a) 30 winning vehicles under OVWSA (b) 40 winning vehicles under OVWSA (c) 50 winning vehicles under OVWSA

(d) 30 winning vehicles under RAWFI (e) 40 winning vehicles under RAWFI (f) 50 winning vehicles under RAWFI

Fig. 6. Under the OVWSA and RAWFI, the number of completion times of sensing tasks in different grid regions changes with the change of the winning vehicle.

of the set of winning opportunistic vehicles selected by this method. In particular, Random has the worst effect, mainly because random selection cannot bring any optimization. The RAWFI result is close to OVWSA because RAWFI also selects the optimal vehicle set through the winner selection method. However, since RAWFI is the optimal solution without considering sensing fairness, this verifies the superiority of the proposed WSA method in terms of sensing fairness.

Fig. 5(b) compares the variation of the CSP overhead with the unit cost of performing sensing tasks for opportunistic vehicles. Here, the candidate vehicle set is set to 60. It can be found that OVWSA is slightly better than RAWFI and TG, and significantly better than Random and CG. This is because CG only selects the opportunistic vehicle with the lowest sensing cost and does not consider the number of tasks that the vehicle can complete, resulting in a lower fairness index and higher overhead. In addition, as the execution cost of sensing tasks increases, the overhead of CSP also gradually increases.

Fig. 6 compares the performance of OVWSA and RAWFI in terms of sensing fairness as the number of winning vehicles changes, with the grid accuracy set to 0.01. Fig. 6(a), (b), and (c) shows the situation of 30, 40, and 50 winning vehicles completing the sensing task under the OVWSA respectively. Correspondingly, Fig. 6(d), (e), and (f) shows the situation of 30, 40, and 50 winning vehicles completing the sensing task under the RAWFI respectively. As the number of winning vehicles increases, the number of sensing tasks completed also increases. It can be clearly seen that in the case of the same number of winning vehicles, the fairness of the sensing task completed by OVWSA is significantly better than RAWFI. This is because the colors of grids in different sub-regions are more balanced, which means that the number of completed sensing tasks in each sub-region grid is close, thus representing better sensing fairness. In addition, the gap in sensing fairness between OVWSA and RAWFI becomes more obvious as the number of winning vehicles increases.

TABLE II
RELATED RESULTS FOR OVRPA

Winner Vehicle ID	Unit Cost	Unit Payment	CSP overhead reduction amount	Number of completed tasks
370	0.140326	1.233298	26.23132	24
179	0.152087	1.648364	35.91063	24
174	0.208489	0.918434	14.90886	21
290	0.14051	0.641551	17.03537	34
14	0.152216	1.260109	37.66836	34
339	0.202298	0.412341	6.721365	32
19	0.114158	0.350992	11.36804	48
278	0.299363	1.869238	31.39749	20
349	0.212298	1.412341	16.72136	32
30	0.121529	0.394767	15.5746	57

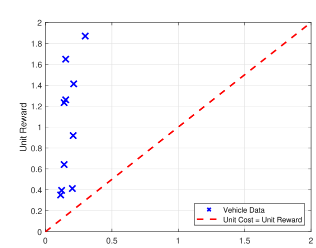


Fig. 7. The validation of OVRPA effectiveness.

We verify the effectiveness of Algorithm 2 and select the results of ten winning opportunistic vehicles for analysis, as shown in Table II and Fig. 7. In Fig. 7, the vehicle data points are represented as (unit cost, unit reward), and it can be seen that the vehicle data are all in the upper left part of the figure. This indicates that the unit reward for each winning opportunistic

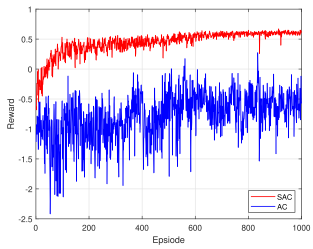
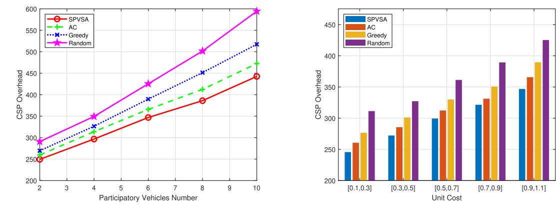
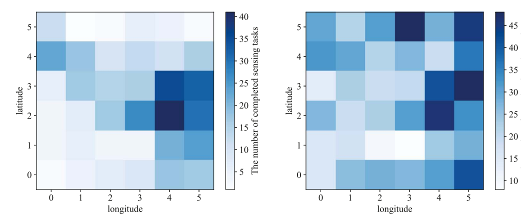


Fig. 8. The average reward trends for each episode.



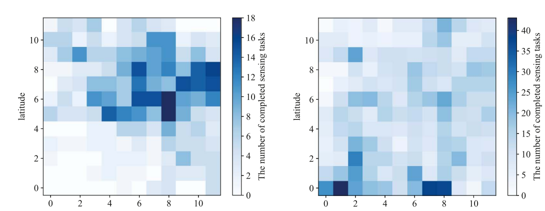
(a) The number of participatory vehicles. (b) The unit cost of performing sensing tasks.

Fig. 9. The CSP overhead changes under different participatory vehicle scenarios.



(a) 20 winning opportunistic vehicles under OVWSA (b) 10 participatory vehicles under SPVSA

Fig. 10. The comparison of sensing fairness in the two-stage sensing process under the $6 * 6$ sub-region grids.



(a) 20 winning opportunistic vehicles under OVWSA (b) 10 participatory vehicles under SPVSA

Fig. 11. The comparison of sensing fairness in the two-stage sensing process under the $12 * 12$ sub-region grids.

vehicle is greater than its unit cost, as expected. In addition, according to Table II, it can be seen that the unit reward of the winning vehicle is related to the degree of reduction in CSP overhead. The greater the contribution, the higher the unit reward received. In contrast, the amount of sensing task completion is not directly related to unit reward. This is because although the vehicle completes more sensing tasks, it does not bring significant improvement to the fairness index, so it cannot obtain higher rewards.

C. Analysis of Participatory Vehicle Experiment Results

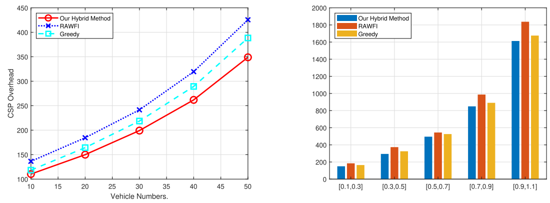
This subsection analyzes the performance of our proposed SPVSA. Please note that this belongs to the second-stage sensing process, upon the completeness of the first-stage opportunistic vehicle sensing.

First, a convergence analysis is performed. We select 6 vehicles to conduct experiments in a $6 * 6$ sub-region grids and based on the results of the 20 winning vehicles in the first stage. Fig. 8 plots a total of 1000 episodes of cumulative average reward convergence curve graph, which is the result of every ten data fittings. As predicted, compared with the AC algorithm, the SPVSA shows faster convergence speed and better performance. It can be seen that as the number of episode training increases, the average reward of SPVSA gradually increases and gradually converges to a relatively stable value. This is because SPVSA maximizes entropy during the optimization process of expected rewards, thereby promoting wider exploration and accelerating the discovery of the best strategy. In addition, it can be seen that the convergence speed of AC is very slow and the fluctuation is large. In summary, after nearly 2000 episodes, the results of AC gradually approached SPVSA, but the fluctuations were still large. Therefore, SPVSA has better convergence performance than AC.

Second, Fig. 9(a) compares the change of CSP overhead with the number of participatory vehicles. Here, the number of participatory vehicles is set between [2, 10], which is also based on the results of the 20 winning opportunistic vehicles in the first stage. It can be seen that as the number of vehicles increases, the cost of CSP gradually increases. This is because more participatory vehicles participate in the sensing task, and CSP must pay more compensation, which leads to an increase in overhead. Importantly, our proposed SPVSA has the lowest overhead, thus verifying the superiority of this method. In comparison, Random has the worst effect, mainly because it cannot bring any optimization in terms of sensing fairness. Although AC is close to our proposed SPVSA, its convergence speed is slow and fluctuates greatly. In addition, Greedy obtains vehicle actions in a greedy manner, which can easily lead to locally optimal solutions, so the CSP overhead is also large.

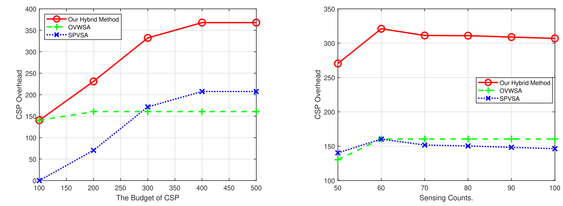
Third, Fig. 9(b) compares the variation of CSP overhead with the unit cost of participatory vehicles performing sensing tasks. This is based on the results of the 20 winning vehicles in the first stage. It can be found that our proposed SPVSA is better than the AC, and significantly better than Random and Greedy. This gap becomes more significant as the unit cost increases. In addition, as the execution cost of sensing tasks increases, the overhead of CSP also gradually increases.

Finally, we analyze the change in sensing fairness after the second stage of the sensing process. Fig. 10 shows the comparison of the sensing fairness of the two-stage sensing process under the 6*6 sub-region grids. At this time, the number of sensing tasks completed by participatory vehicles is set to 50. Similarly, Fig. 11 shows the comparison of the sensing fairness of the two-stage sensing process under the 12*12 sub-region grids. At



(a) The CSP overhead changes with vehicles number. (b) The CSP overhead changes with task cost.

Fig. 12. The comparison of our two-stage hybrid sensing method with the single-stage method.



(a) The CSP overhead changes with budget. (b) The CSP overhead changes with sensing counts.

Fig. 13. The comparison of related parameters affecting two-stage hybrid sensing methods.

this time, the number of sensing tasks completed by participatory vehicles is set to 100. Here, after the first stage of sensing process, Figs. 10(a) and 11(a) show the number of completed sensing tasks in each sub-region grid. Based on the first-stage sensing results, the second-stage sensing process is performed. Figs. 10(b) and 11(b) show the second-stage sensing results, that is, the number of completed sensing tasks in each sub-region grid. Compared with the results of the first stage, it can be clearly seen that the sensing fairness is significantly improved in the second stage. This proves the effectiveness of our proposed SPVSA, which effectively improves the sensing fairness in the second-stage sensing process. In particular, in Fig. 11(a), there are fewer sensing tasks completed in the lower left sub-region grids. But in Fig. 11(b), the number of completed sensing tasks in the lower left sub-region grids increases significantly. This clearly reflects the efforts of our proposed SPVSA method in terms of sensing fairness.

D. Analysis of Two-Stage Relationship

First, we compare the proposed two-stage hybrid sensing method with methods involving only opportunistic or participatory vehicles. Here, we use RAWFI [16] to represent the comparison scheme in which only opportunistic vehicles participate, and Greedy [10] to represent the comparison scheme in which only participating vehicles participate. It should be noted that we have specifically adjusted the number of vehicles in RAWFI and the task completion cost in Greedy. This ensures that the results of the comparison scheme and the two-stage method proposed in this paper remain on the same order of magnitude. Here, the ratio of opportunistic vehicles and participatory vehicles in our proposed two-stage hybrid sensing method is 1:9. Fig. 12(a) shows the variation of CSP overhead with the number of vehicles. Fig. 12(b) shows the variation of CSP overhead with the task cost when a total of 20 vehicles are involved. It can be seen that our proposed two-stage hybrid sensing method outperforms Greedy and significantly outperforms RAWFI. Here, Greedy can better explore different sub-region grids to complete the task to maintain sensing fairness, but it is easy to fall into the local optimal solution. Since RAWFI does not consider sensing fairness and only selects the winning vehicle with the most appropriate cost, it has a disadvantage in sensing fairness and results in poorer results.

Second, we analyze the relationship between the two stage sensing methods. In our paper, the relevant parameters are the CSP budget B and the number of sensing counts η for each sub-region grid. Here, we set up 40 opportunistic vehicles and 4 participatory vehicles to participate in the sensing task. Fig. 13(a) shows the changes in CSP overhead under different budgets. It can be seen that when the budget is low, the sensing demands of OVWSA in the first-stage cannot be met, so the result of SPVSA is 0. As the budget increases, the OVWSA is completed, the SPVSA begins, and until the SPVSA is completed, there is still budget left. Fig. 13(b) shows the changes of CSP overhead with the number of sensing counts of the sub-region grid. Here we consider the 6*6 grid region and give sufficient budget to ensure that the two-stage sensing process can be completed. It can be seen that the CSP overhead under OVWSA increased slightly at the beginning and remained unchanged. This is because the maximum number of sensing counts of the opportunistic vehicle in the sub-region grid in this state is 58, so increasing η to more than 60 counts has no impact on the first-stage. In the second-stage, SPVSA continues its efforts to maintain sensing fairness. It can be seen that when the first-stage sensing process is not stable, the overhead of CSP is also increasing. When the first-stage sensing process has no impact, the overhead of CSP is reduced, which shows that SPVSA has made efforts in sensing fairness. However, due to the small size of our scenario setting, this change is not obvious. As the scenario setting expands, the effect of the second-stage sensing process becomes more obvious.

Overall, we verify the effectiveness of the proposed two-stage hybrid sensing method. This method makes an effort in sensing fairness by taking advantage of the mobility characteristics of both types of vehicles. In particular, it makes the best trade-off in terms of CSP overhead and sensing fairness, achieving a win-win situation. Compared with the current single type of sensing method, the advantages are obvious. In future work, we will further explore the relationship between the two stages, as well as the interaction between them.

VII. Conclusion

This paper proposed a two-stage hybrid sensing method applied to vehicular crowdsensing. First, in the first stage of the sensing process, a reverse auction-based incentive mechanism was proposed to select the lowest-cost opportunistic vehicle to complete the sensing task. Second, in the second stage of the sensing process, based on the sensing results of the first stage, a SAC-based participatory vehicle scheduling method was proposed. In addition, this paper also considered the fairness issue of sensing task completion and introduced a sensing fairness index. Finally, extensive evaluations based on real-world datasets show that the proposed method works well and outperforms other baseline schemes.

References

[1] X. Li, G. Feng, Y. Liu, S. Qin, and Z. Zhang, "Joint sensing, communication, and computation in mobile crowdsensing enabled edge networks," *IEEE Trans. Wirel. Commun.*, vol. 22, no. 4, pp. 2818–2832, Apr. 2023.

[2] H. Zhou, Z. Wang, N. Cheng, D. Zeng, and P. Fan, "Stackelberg-game-based computation offloading method in cloud–edge computing networks," *IEEE Internet Things J.*, vol. 9, no. 17, pp. 16 510–16 520, Sep. 2022.

[3] K. Liu, S. Peng, W. Gong, B. Zhang, and C. Li, "Hybrid user based task assignment for mobile crowdsensing: Problem and algorithm," *IEEE Internet Things J.*, vol. 11, no. 11, pp. 19589–19601, Jun. 2024, doi: 10.1109/JIOT.2024.3367958.

[4] Y. Liu, L. Kong, and G. Chen, "Data-oriented mobile crowdsensing: A comprehensive survey," *IEEE Commun. Surv. Tutor.*, vol. 21, no. 3, pp. 2849–2885, Third Quarter, 2019.

[5] X. Yang, B. Gu, B. Zheng, B. Ding, Y. Han, and K. Yu, "Toward incentive-compatible vehicular crowdsensing: An edge-assisted hierarchical framework," *IEEE Netw.*, vol. 36, no. 2, pp. 162–167, Mar./Apr. 2022.

[6] L. Liu, L. Wang, Z. Lu, Y. Liu, W. Jing, and X. Wen, "Cost-and-quality aware data collection for edge-assisted vehicular crowdsensing," *IEEE Trans. Veh. Technol.*, vol. 71, no. 5, pp. 5371–5386, May 2022.

[7] Y. Fu, X. Qin, X. Zhang, and Y. Jia, "Hybrid recruitment scheme based on deep learning in vehicular crowdsensing," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 10, pp. 10 735–10 748, Oct. 2023.

[8] J. Wang et al., "HyTasker: Hybrid task allocation in mobile crowd sensing," *IEEE Trans. Mob. Comput.*, vol. 19, no. 3, pp. 598–611, Mar. 2020.

[9] Y. Ren, H. Jiang, X. Feng, Y. Zhao, R. Liu, and H. Yu, "ACP-based modeling of the parallel vehicular crowd sensing system: Framework, components and an application example," *IEEE Trans. Intell. Veh.*, vol. 8, no. 2, pp. 1536–1548, Feb. 2023.

[10] H. Jiang, Y. Ren, J. Fang, Y. Yang, L. Xu, and H. Yu, "SHIP: A state-aware hybrid incentive program for urban crowd sensing with for-hire vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 3, pp. 3041–3053, Mar. 2024.

[11] F. Yucel and E. Bulut, "Online stable task assignment in opportunistic mobile crowdsensing with uncertain trajectories," *IEEE Internet Things J.*, vol. 9, no. 11, pp. 9086–9101, Jun. 2022.

[12] H. Zhou, Z. Wang, G. Min, and H. Zhang, "UAV-aided computation offloading in mobile-edge computing networks: A stackelberg game approach," *IEEE Internet Things J.*, vol. 10, no. 8, pp. 6622–6633, Apr. 2023.

[13] E. Wang, H. Wang, Y. Yang, and W. Liu, "Truthful incentive mechanism for budget-constrained online user selection in mobile crowdsensing," *IEEE Trans. Mob. Comput.*, vol. 21, no. 12, pp. 4642–4655, Dec. 2022.

[14] H. Zhou, Z. Wang, H. Zheng, S. He, and M. Dong, "Cost minimization-oriented computation offloading and service caching in mobile cloud-edge computing: An A3C-based approach," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 3, pp. 1326–1338, May/Jun. 2023.

[15] Y. Xu, Y. Wang, J. Ma, and Q. Jin, "PSARE: A RL-based online participant selection scheme incorporating area coverage ratio and degree in mobile crowdsensing," *IEEE Trans. Veh. Technol.*, vol. 71, no. 10, pp. 10 923–10 933, Oct. 2022.

[16] G. Fan et al., "Joint scheduling and incentive mechanism for spatio-temporal vehicular crowd sensing," *IEEE Trans. Mob. Comput.*, vol. 20, no. 4, pp. 1449–1464, Apr. 2021.

[17] M. Hu, Z. Zhong, Y. Niu, and M. Ni, "Duration-variable participant recruitment for urban crowdsourcing with indeterministic trajectories," *IEEE Trans. Veh. Technol.*, vol. 66, no. 11, pp. 10 271–10 282, Nov. 2017.

[18] Y. Lai, Y. Xu, D. Mai, Y. Fan, and F. Yang, "Optimized large-scale road sensing through crowdsourced vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 4, pp. 3878–3889, Apr. 2022.

[19] H. Zhou, T. Wu, X. Chen, S. He, D. Guo, and J. Wu, "Reverse auction-based computation offloading and resource allocation in mobile cloud-edge computing," *IEEE Trans. Mob. Comput.*, vol. 22, no. 10, pp. 6144–6159, Oct. 2023.

[20] G. Gao, M. Xiao, J. Wu, L. Huang, and C. Hu, "Truthful incentive mechanism for nondeterministic crowdsensing with vehicles," *IEEE Trans. Mob. Comput.*, vol. 17, no. 12, pp. 2982–2997, Dec. 2018.

[21] X. Chen, L. Zhang, Y. Pang, B. Lin, and Y. Fang, "Timeliness-aware incentive mechanism for vehicular crowdsourcing in smart cities," *IEEE Trans. Mob. Comput.*, vol. 21, no. 9, pp. 3373–3387, Sep. 2022.

[22] Y. Zhao and C. H. Liu, "Social-aware incentive mechanism for vehicular crowdsensing by deep reinforcement learning," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 4, pp. 2314–2325, Apr. 2021.

[23] C. Xu and W. Song, "Intelligent task allocation for mobile crowdsensing with graph attention network and deep reinforcement learning," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 2, pp. 1032–1048, Mar./Apr. 2023.

[24] H. Zhou, K. Jiang, S. He, G. Min, and J. Wu, "Distributed deep multi-agent reinforcement learning for cooperative edge caching in Internet-of-Vehicles," *IEEE Trans. Wirel. Commun.*, vol. 22, no. 12, pp. 9595–9609, Dec. 2023.

[25] R. Ding, Z. Yang, Y. Wei, H. Jin, and X. Wang, "Multi-agent reinforcement learning for urban crowd sensing with for-hire vehicles," in *Proc. IEEE IEEE Conf. Comput. Commun.*, 2021, pp. 1–10.

[26] C. H. Liu, Z. Dai, Y. Zhao, J. Crowcroft, D. Wu, and K. K. Leung, "Distributed and energy-efficient mobile crowdsensing with charging stations by deep reinforcement learning," *IEEE Trans. Mob. Comput.*, vol. 20, no. 1, pp. 130–146, Jan. 2021.

[27] L. Ding, D. Zhao, M. Cao, and H. Ma, "When crowdsourcing meets unmanned vehicles: Toward cost-effective collaborative urban sensing via deep reinforcement learning," *IEEE Internet Things J.*, vol. 8, no. 15, pp. 12 150–12 162, Aug. 2021.

[28] C. H. Liu, Z. Chen, and Y. Zhan, "Energy-efficient distributed mobile crowd sensing: A deep learning approach," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 6, pp. 1262–1276, Jun. 2019.

[29] J. Shi, J. Du, J. Wang, J. Wang, and J. Yuan, "Priority-aware task offloading in vehicular fog computing based on deep reinforcement learning," *IEEE Trans. Veh. Technol.*, vol. 69, no. 12, pp. 16 067–16 081, Dec. 2020.

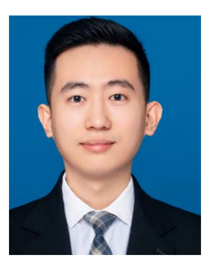
[30] X. Fan, M. Liu, Y. Chen, S. Sun, Z. Li, and X. Guo, "RIS-assisted UAV for fresh data collection in 3D urban environments: A deep reinforcement learning approach," *IEEE Trans. Veh. Technol.*, vol. 72, no. 1, pp. 632–647, Jan. 2023.

[31] W. Wang, X. Zhou, T. Qiu, X. He, and S. Ge, "Location-privacy-aware service migration against inference attacks in multiuser MEC systems," *IEEE Internet Things J.*, vol. 11, no. 1, pp. 1413–1426, Jan. 2024.

[32] X. Cao et al., "Trajectory penetration characterization for efficient vehicle selection in HD map crowdsourcing," *IEEE Internet Things J.*, vol. 8, no. 6, pp. 4526–4539, Mar. 2021.

[33] Q. Xu, Z. Su, D. Fang, and Y. Wu, "BASIC: Distributed task assignment with auction incentive in UAV-enabled crowdsensing system," *IEEE Trans. Veh. Technol.*, vol. 73, no. 2, pp. 2416–2430, Feb. 2024.

[34] N. Parvaresh and B. Kantarci, "A continuous actor-critic deep Q-learning-enabled deployment of UAV base stations: Toward 6G small cells in the skies of smart cities," *IEEE Open J. Commun. Soc.*, vol. 4, pp. 700–712, 2023.



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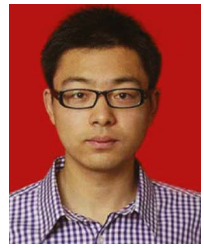

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